

No.	Answer	Solution
28	B	Transition from $n = 2$ to $n = 1$, $E_2 - E_1 = \frac{hc}{\lambda_1} \text{ ---- (1) (where } \lambda_1 = 440 \text{ nm)}$ Transition from $n = 3$ to $n = 2$ produces electromagnetic radiation of longer wavelength, i.e. 590 nm. $E_3 - E_2 = \frac{hc}{\lambda_2} \text{ ---- (2) (where } \lambda_2 = 590 \text{ nm)}$ Third transition is transition from $n = 3$ to $n = 1$, $E_3 - E_1 = \frac{hc}{\lambda_3} \text{ ---- (3)}$ (1) + (2) = (3), $\frac{hc}{\lambda_1} + \frac{hc}{\lambda_2} = \frac{hc}{\lambda_3}$ $\frac{1}{\lambda_1} + \frac{1}{\lambda_2} = \frac{1}{\lambda_3}$ $\frac{1}{440} + \frac{1}{590} = \frac{1}{\lambda_3}$ $\lambda_3 = \left(\frac{1}{440} + \frac{1}{590} \right)^{-1}$ $= 252$ $\approx 250 \text{ nm}$